UNIT 5 – Evaluation Metrics & Neural Learning

1. Explain confusion matrix in detail.

Definition:
A Confusion Matrix is a table used to evaluate the performance of a classification model by comparing actual values with predicted values.

Structure:

Actual / Predicted	**Positive**	**Negative**
Positive	TP	FN
Negative	FP	TN

Terms:

- **TP (True Positive):** Actual positive and predicted positive
- **TN (True Negative):** Actual negative and predicted negative
- **FP (False Positive):** Actual negative but predicted positive
- **FN (False Negative):** Actual positive but predicted negative

Importance:
It helps calculate:

- Accuracy
- Precision
- Recall
- F1 Score

Example:
Disease detection system me confusion matrix batata hai kitne patients correctly detect hue aur kitne miss hue.

2. Define and Explain: Accuracy, Precision, Recall, F1 Score.

Accuracy
Correct predictions / Total predictions

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Measures overall correctness.

Precision
Correct positive predictions / Total predicted positives

$$Precision = \frac{TP}{TP + FP}$$

Measures quality of positive predictions.

Recall
Correct positive predictions / Total actual positives

$$Recall = \frac{TP}{TP + FN}$$

Measures ability to detect positives.

F1 Score

Harmonic mean of Precision and Recall

$$F1 = \frac{2PR}{P + R}$$

Used when both precision and recall are important.

3. Explain ROC curve and AUC.

ROC Curve:

Receiver Operating Characteristic Curve is a graph used to evaluate a classifier by plotting:

- X-axis = False Positive Rate
- Y-axis = True Positive Rate

A better model curve stays near top-left corner.

AUC:

Area Under Curve

- 1 = Perfect model
- 0.5 = Random model
- Higher AUC = Better classifier

Uses:

Fraud detection, disease diagnosis, spam filtering.

4. What are significance tests in ML evaluation.

Significance tests are statistical methods used to determine whether observed performance differences between models are real or happened by chance.

Purpose:

- Compare Model A vs Model B
- Check if improvement is meaningful

Common Tests:

- T-Test
- Z-Test
- ANOVA

P-value:

If $p < 0.05$ → Significant result

5. Explain error correction in perceptrons.

A Perceptron is a simple neural network for binary classification.

If predicted output is wrong, weights are updated to reduce future error.

Weight Update Rule:

$$w_{new} = w_{old} + \eta(target - output)x$$

Steps:

1. Input do
2. Output predict karo
3. Compare with actual output
4. Update weights if error exists

This process is called error correction.

6. What is overfitting and underfitting.

Overfitting:

Model training data ko bahut achha learn karta hai but new data pe poor performance deta hai.

Causes:

- Complex model
- Too much training

Underfitting:

Model data ke pattern ko properly learn nahi karta.

Causes:

- Too simple model
- Insufficient training

Solution:

- Cross-validation
- Regularization
- Better model tuning

7. Explain cross-validation techniques.

Cross-validation is used to test model performance on different subsets of data.

Types:

1. Hold-Out Method

Train-test split (e.g., 80%-20%)

2. K-Fold Cross Validation

Data ko K parts me divide karte hain. Har fold test banega ek baar.

3. Stratified K-Fold

Class distribution same rakhta hai.

4. Leave-One-Out

One sample test, rest training.

Advantage:

Better model evaluation.

8. Discuss loss functions used in ML.

Loss function measures prediction error.

Common Loss Functions:

Classification:

- Binary Cross Entropy
- Categorical Cross Entropy
- Hinge Loss

Regression:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

Purpose:

Training ke time model error minimize karta hai.

9. Explain gradient descent in ML.

Gradient Descent is an optimization algorithm used to minimize loss function by updating model parameters step-by-step.

Idea:

Loss kam karna hai by moving in direction of negative gradient.

Update Rule:

$$\theta = \theta - \alpha \nabla J(\theta)$$

Where:

- α = learning rate
- θ = parameter

Types:

- Batch Gradient Descent
- Stochastic Gradient Descent
- Mini-batch Gradient Descent

10. Write short notes on performance evaluation.

Performance evaluation means measuring how well an ML model performs.

For Classification:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

For Regression:

- MAE
- MSE

- R^2 Score
- Adjusted R^2

Importance:

- Select best model
- Compare algorithms
- Improve accuracy
- Detect overfitting

UNIT 6 – Ensemble Learning & Random Forests

1. Define ensemble learning and its techniques. Why is it needed.

Definition:

Ensemble Learning is a machine learning technique in which multiple models are combined to improve prediction accuracy and robustness.

Why Needed?

Single model may suffer from:

- Low accuracy
- Overfitting
- High variance
- Poor generalization

Ensemble methods solve these problems by combining many learners.

Main Techniques:

1. **Bagging** – Parallel models on random subsets
2. **Boosting** – Sequential models correcting mistakes
3. **Stacking** – Multiple models + meta learner

Examples:

Random Forest, AdaBoost, XGBoost

2. Explain Bagging with example.

Definition:

Bagging (Bootstrap Aggregating) trains multiple models on random samples of data with replacement and combines their outputs.

Steps:

1. Create bootstrap samples
2. Train same model on each sample
3. Predict using all models
4. Final output by:
    - Voting (classification)
    - Averaging (regression)

Example:

Random Forest uses many decision trees trained on different samples.
If trees predict:

- Tree1 = Apple
- Tree2 = Apple
- Tree3 = Banana

Final = Apple (majority vote)

3. Describe Boosting technique.

Definition:
Boosting is an ensemble method where weak learners are trained sequentially, and each new model focuses on mistakes of the previous model.
Working:

1. Train first weak model
2. Find wrong predictions
3. Increase focus on wrong samples
4. Train next model
5. Combine all learners

Goal:

- Reduce bias
- Increase accuracy

Examples:
AdaBoost, Gradient Boosting, XGBoost

4. Explain AdaBoost algorithm step by step.

Full Form:
Adaptive Boosting
Steps:

1. Assign equal weights to all training samples
2. Train first weak learner
3. Identify misclassified samples
4. Increase weights of wrong samples
5. Train next learner on updated data
6. Repeat for many iterations
7. Combine all weak learners using weighted voting

Result:
Many weak learners become one strong model.

5. What is Random Forest. Explain with numerical illustarion.

Definition:
Random Forest is an ensemble algorithm that combines multiple decision trees and gives final output using majority voting or averaging.
Working:

1. Random data samples selected

2. Random features selected
3. Build many trees
4. Combine outputs

Numerical Example:

Suppose 5 trees predict class:

- Tree1 = Yes
- Tree2 = Yes
- Tree3 = No
- Tree4 = Yes
- Tree5 = No

Final Output = **Yes** (3 votes)

For Regression:

Predictions = 10, 12, 14

Average:

$$\frac{10 + 12 + 14}{3} = 12$$

6. Compare Decision Tree and Random Forest.

Feature	Decision Tree	Random Forest
Model Type	Single tree	Many trees
Accuracy	Moderate	Higher
Overfitting	High chance	Low
Speed	Faster	Slower
Stability	Less	More
Interpretation	Easy	Hard

Conclusion:

Random Forest is more powerful and accurate.

7. Explain XGBoost and its advantages.

Full Form:

Extreme Gradient Boosting

Definition:

XGBoost is an advanced boosting algorithm that builds trees sequentially and optimizes errors efficiently.

Advantages:

1. High accuracy
2. Fast training
3. Regularization reduces overfitting

4. Handles missing values
5. Feature importance available
6. Works on large datasets

Uses:
Fraud detection, forecasting, ranking systems.

8. Discuss bias reduction using ensemble methods.

Bias Meaning:
Error due to overly simple assumptions.

How Ensemble Reduces Bias:

Boosting:
Each new model corrects previous mistakes, reducing bias.

Example:
Weak learners individually poor hote hain, but combined strong ban jaate hain.

Algorithms:

- AdaBoost
- Gradient Boosting
- XGBoost

Result:
Better accuracy and learning.

9. Compare Bagging vs Boosting.

Feature	Bagging	Boosting
Training	Parallel	Sequential
Goal	Reduce variance	Reduce bias
Data	Random subsets	Focus on mistakes
Speed	Faster	Slower
Example	Random Forest	AdaBoost

Conclusion:

Bagging prevents overfitting, Boosting improves weak models.

10. Explain real-world applications of ensemble learning.

1. Banking

- Loan default prediction
- Fraud detection

2. Healthcare

- Disease diagnosis
- Cancer detection

3. E-commerce

- Customer churn prediction
- Recommendation systems

4. Finance

- Stock price prediction
- Risk analysis

5. Self-Driving Cars

- Object recognition

6. Search Engines

- Ranking pages